

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic
Medicine

Pathology: Restrictive and Miscellaneous Cardiomyopathies

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Session Objectives

1. Discuss, describe, and explain the **general** characteristics and **pathophysiology** of **restrictive cardiomyopathies**.
2. Discuss, describe, and explain the **general** characteristics and **pathophysiology** of **cardiac amyloidosis**.
3. Briefly describe **amyloid pathology**.
4. Discuss and describe miscellaneous restrictive cardiomyopathies including **endomyocardial fibrosis**, **Loeffler endomyocarditis**, and **endocardial fibroelastosis**.
5. Summarize and **compare** dilated, hypertrophic, and restrictive **cardiomyopathies**.

Restrictive Cardiomyopathy

- **Primary** ↓ in ventricular **compliance**
 - Results in **impaired** ventricular filling during **diastole**
 - Abnormally **rigid**, NOT necessarily thickened
 - Contractile (**systolic**) function of LV usually **not affected**
- Types: **Idiopathic**, or associated with diseases such as **radiation fibrosis**, **amyloidosis**, **sarcoidosis**, metastatic **tumor**, or products of inborn **errors of metabolism**:
 - Main types & causes: Grouped by what causes muscle to get stiff
 - Infiltrative diseases: **Amyloidosis (most common)**, **sarcoidosis**: causes patches of inflammation and scarring
 - Storage diseases: **Hemochromatosis**: Causes too much iron in, **Fabry disease** (rare genetic **fat**-storage disorder)
 - Endomyocardial disorders: **Endomyocardial fibrosis** (common in **tropical** regions & developing countries, **hyper-eosinophilic syndrome (Loeffler endomyocarditis** - damage from high levels of eosinophils)
 - Non-infiltrative & idiopathic: **Idiopathic** (exact cause is unknown), **scleroderma** (systemic sclerosis that affects connective tissue)

Source: Robbins, Cotran & Kumar Pathologic Basis of Disease, 11th ed., 2026, Ch. 12

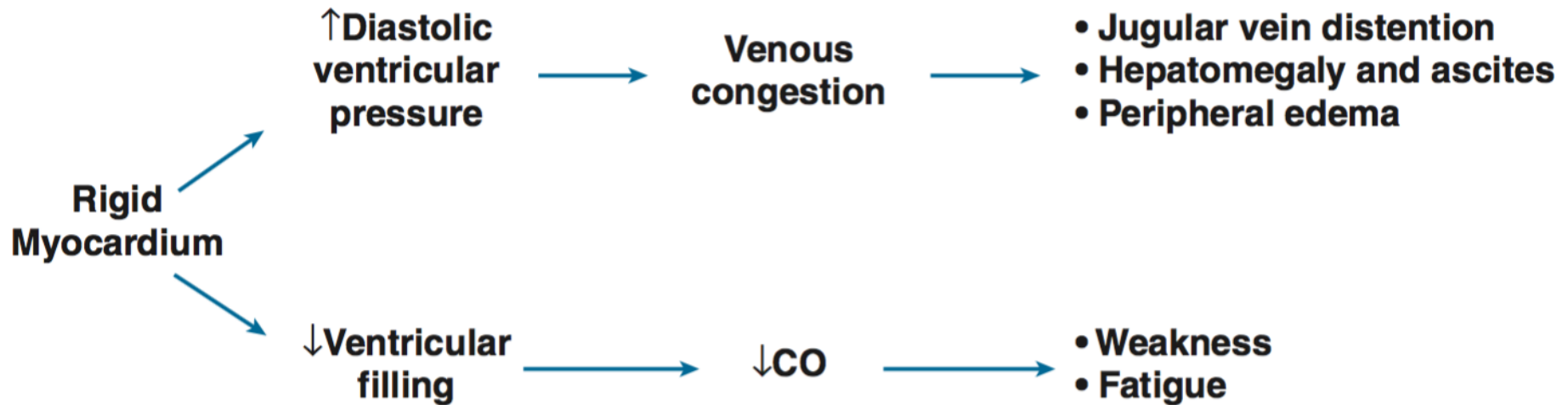
Idiopathic Restrictive Cardiomyopathy

- **Ventricles** are normal size or slightly enlarged, not dilated, myocardium is firm
- **Biatrial dilation** – can cause Sx of left and right heart failure & arrhythmias
- Histopathology: Patchy or **diffuse interstitial fibrosis** varying from minimal to severe
 - Endomyocardial **biopsy** can show disease specific findings in some cases



<https://www.studyblue.com/notes/note/n/week-82-myocarditis-and-cardiomyopathies-/deck/9469649>

Pathophysiology



Lilly, Pathophysiology of Heart Disease, 5th edition, 2011, p. 258

Clinical presentation

- Signs and symptoms of **CHF** are common such as:
 - ↑ Ventricular diastolic **filling pressure**
 - ↓ **CO** (**fatigue** & decreased exercise tolerance)
 - Systemic **congestion** (often more prominent than pulmonary) → **JVD**, peripheral **edema** & **ascites** with **hepatomegaly**, tender liver, pulmonary **rales**
- CXR: **Normal-sized heart** with signs of **pulmonary congestion**
- ECG: **Nonspecific** ST & T wave abnormalities; conduction disturbances such as AV block or BBB
- Restrictive cardiomyopathies share nearly **identical symptoms**, physical **signs**, and hemodynamic profiles with **constrictive pericarditis** - distinguishable because constrictive pericarditis is often correctable

Amyloid

- Abnormal proteins deposits build up in tissue - **amyloid fibrils** are nonbranching and form protein aggregates w/ **beta-pleated sheet** structure of protein (this conformation accounts for its staining properties)
 - Beta-sheet form of amyloid is **proteolysis-resistant**
- There are many different **types** of amyloid, each due to a specific protein **misfolding** due to genetic effects, and/or acquired environmental factors:
 - Grouped into **localized** (primarily a single organ, caused by several different amyloids) & **systemic** forms, a few can identify as either
 - Sometimes classified as **primary** where there is no other disease that is causing the condition (usually affects heart, kidneys, liver, nerves)
 - **Secondary** usually triggered by inflammatory disease, like rheumatoid arthritis, most commonly affects kidneys, liver, spleen
 - **Four most common types of systemic amyloidosis:**
 - Light chain (**AL**)-**multiple myeloma**
 - **Chronic inflammation (AA)** – ex. rheumatoid arthritis
 - **Dialysis-related (A β 2M)**
 - Hereditary & old age/**senile (ATTR, wild-type transthyretin** amyloid overproduction)

Amyloidosis - S/S & Dx

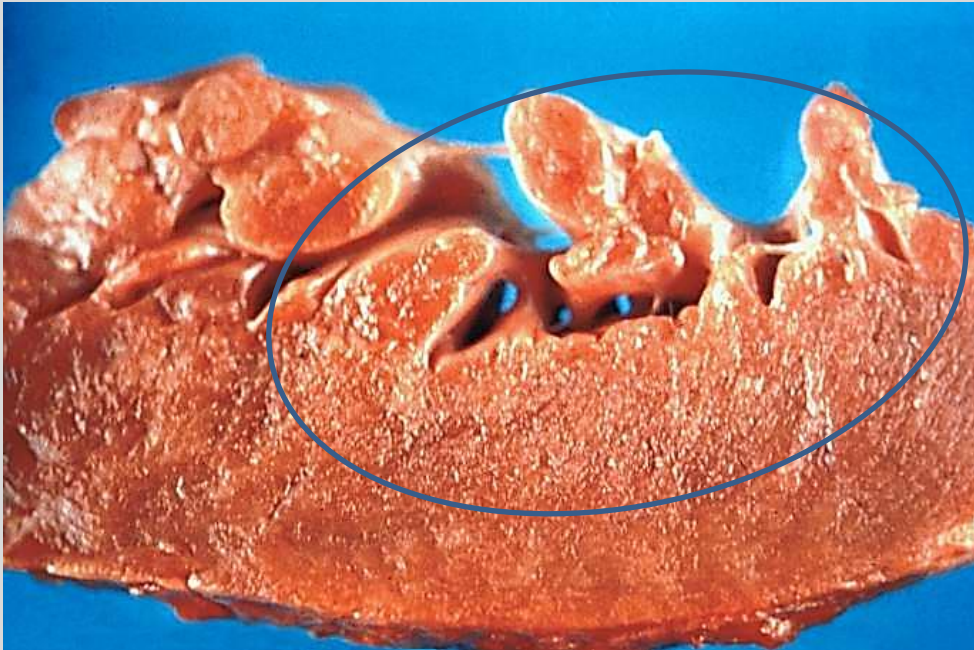
- Non-specific & **vague signs & symptoms w/ varied presentations**
 - Fatigue
 - Peripheral edema
 - Shortness of breath
 - Palpitations
 - Orthostatic hypotension
- In **AL amyloidosis**, may include enlargement of the **tongue** and **periorbital** purpura
- In **wild-type ATTR amyloidosis**, may include heart & tendons: Bilateral **carpal tunnel** syndrome, lumbar **spinal stenosis**, biceps tendon rupture, small fiber **neuropathy**, and **autonomic dysfunction**
- Diagnosis may be **suspected when** protein is found in **urine**, **organ** enlargement, or unexplained problems w/ multiple peripheral **nerves** - diagnosis confirmed by tissue **biopsy**

Cardiac amyloidosis

- Major organ involved in **senile amyloidosis**
- Amyloid deposits in **atria & ventricles** or atria alone
 - If subendocardial, **conduction system** may be disturbed
 - **Isolated atrial** amyloidosis is 4X more common in **African Americans > 60 years** because of gene mutation in 4% of African American population producing mutant form of transthyretin
- **Grossly** may show no significant findings
 - May be mildly enlarged & firm (**typical restrictive cardiomyopathy**), rubbery, non-compliant
 - May see waxy semi-translucent nodules (subtle) especially on left atrial endocardium
- **Histopathology: Hyaline** eosinophilic **deposits** of amyloid may be found in the interstitium, conduction tissue, valves, endocardium, pericardium & small intramural coronary arteries
 - With H&E stained tissue sections, **amorphous, eosinophilic**
 - **Congo Red** stain shows salmon pink deposits w/ **apple green** birefringence upon polarization microscopy

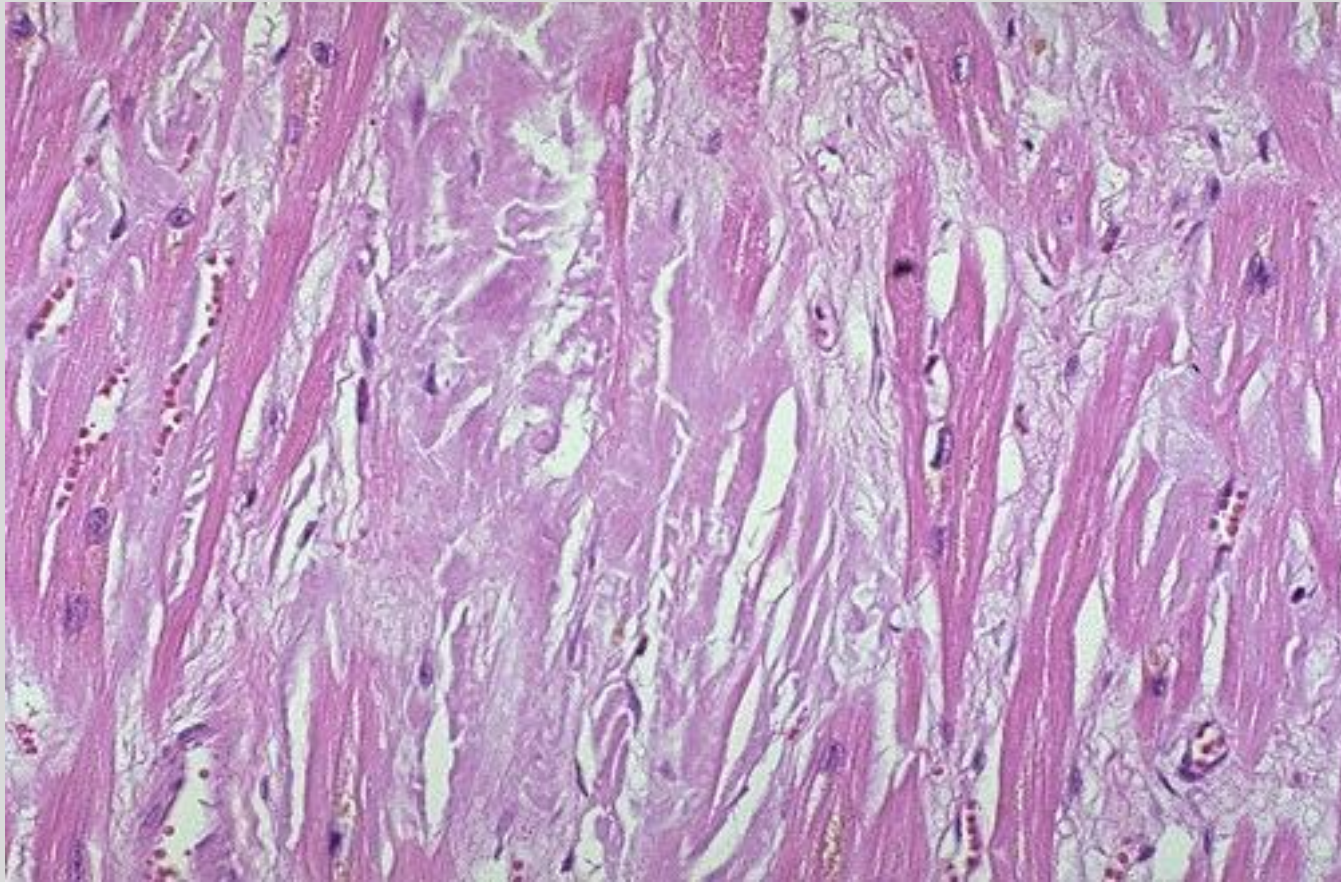
Cardiac amyloidosis pathology

- Amyloid deposits give the myocardium a **pale**, translucent, **waxy** appearance
- Rubbery, non-compliant



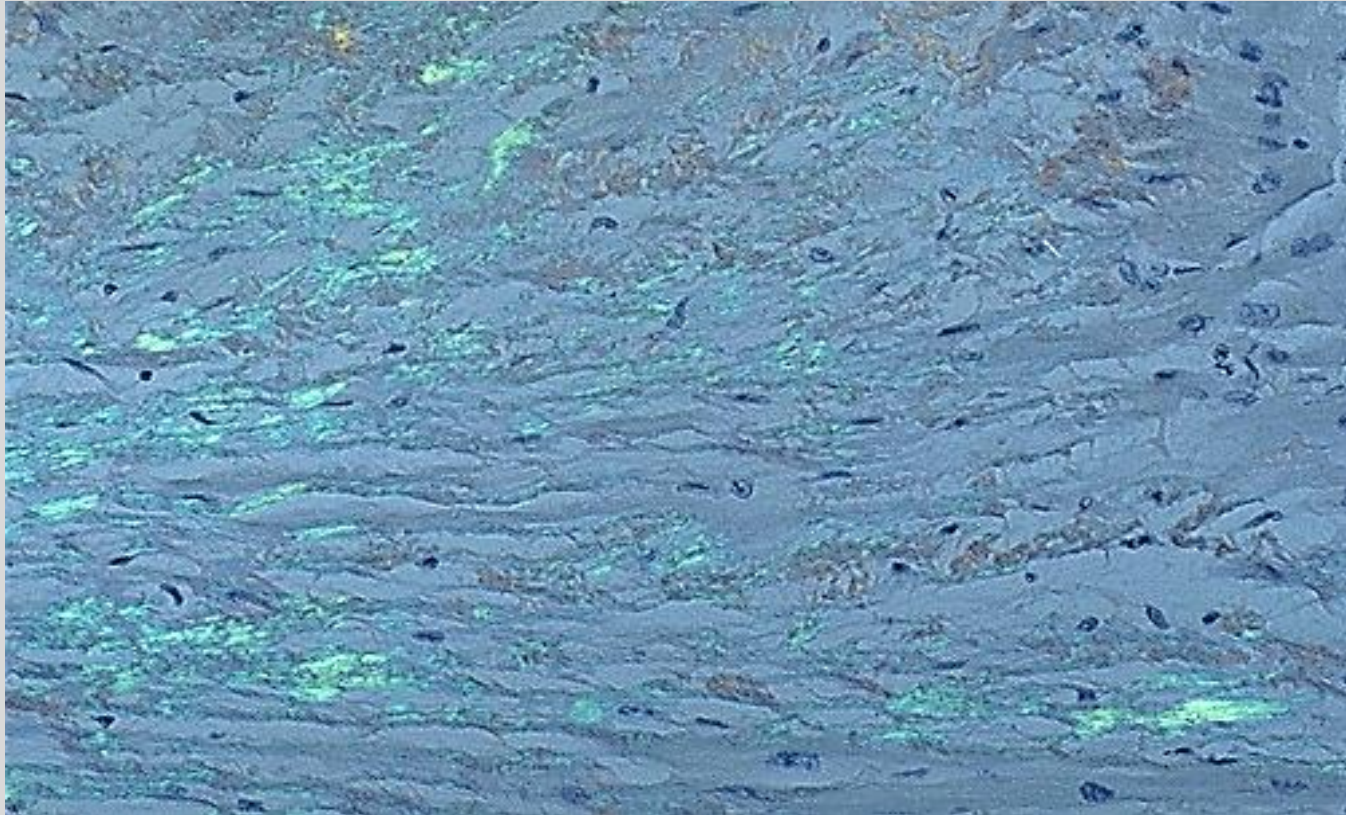
http://www.wikidoc.org/index.php/File:Cardiac_amyloidosis_case_1.jpg

Cardiac amyloidosis histopathology: Amorphous deposits of **pale pink** material between myocardial fibers, characteristic for amyloid, one cause for "infiltrative" or "restrictive" cardiomyopathy



H&E stain

Congo red stain: Light microscopy shows amyloid stains orange-red/salmon pink, but **w/ polarized light**, the amyloid has an "apple-green" birefringence



<http://library.med.utah.edu/WebPath/CVHTML/CV073.html>

Other restrictive cardiomyopathies:

Endomyocardial fibrosis

- AKA tropical endomyocardial fibrosis
- Usually a disease of **children & young** adults especially in **Africa/tropics**, rare in North America
- **Fibrosis** of ventricular **endocardium & subendocardium** extending from **apex to AV valves** (may involve valves)
- Volume & **compliance** of ventricles ↓
- Ventricular mural **thrombi** may develop (endocardial fibrosis may result from thrombus organization)
- Unknown etiology & pathogenesis; however, because pathology **resembles eosinophilic cardiomyopathy** sometimes considered part of a single disease process that also **includes Loeffler endocarditis** (eosinophilic endomyocardial fibrosis)

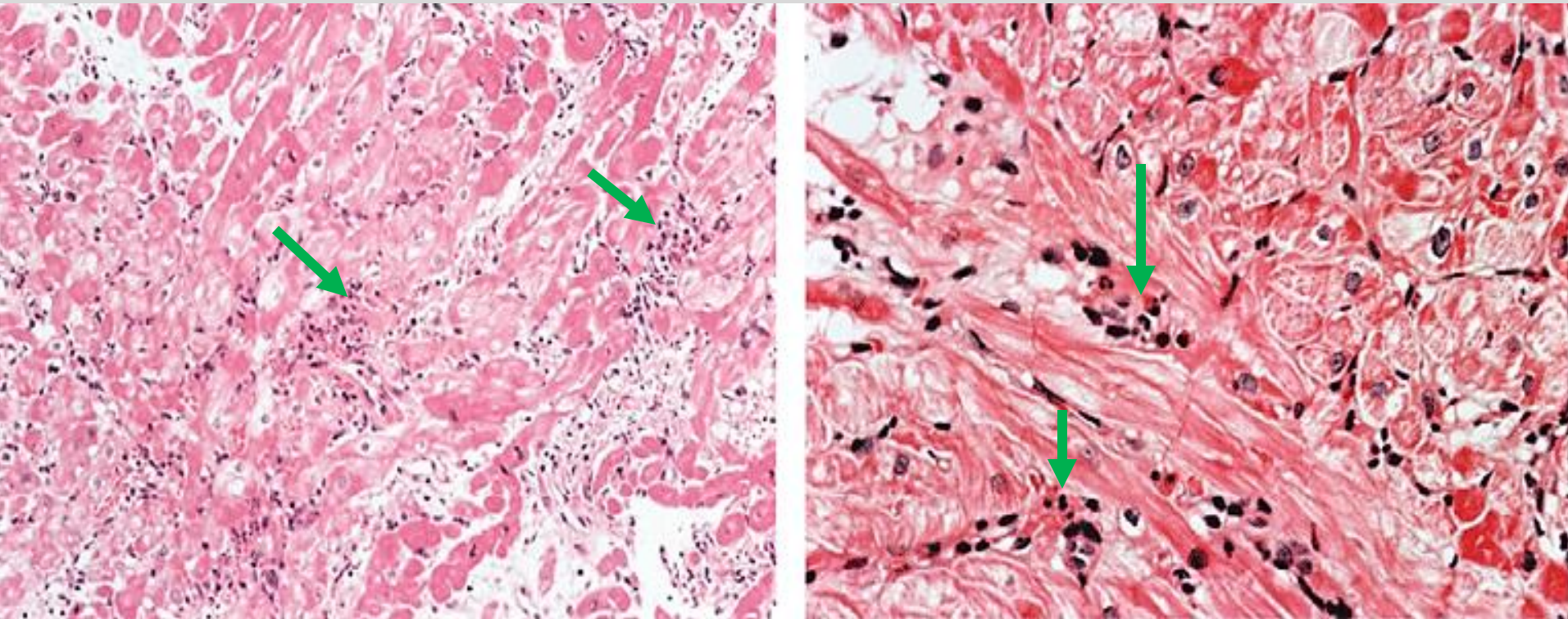
Source: Robbins, Cotran & Kumar Pathologic Basis of Disease, 11th ed., 2026, Ch. 12

Loeffler endomyocarditis

- **Rare**, restrictive cardiomyopathy w/ **endomyocardial fibrosis** & usually large **mural thrombi**
- ↑ **peripheral blood eosinophils** & eosinophils in multiple organs including **heart**
- **Abnormal eosinophils** may release toxic products like major basic protein which cause endocardial damage → areas of necrosis with eosinophilic infiltrate → scarring → layering of endocardium by thrombus → formation of **thrombus**
- Associated w/ PDGFR alpha/beta gene abnormalities - seen in chronic myeloproliferative disorders that produce hypereosinophilia
- **Poor prognosis**

Source: Robbins, Cotran & Kumar Pathologic Basis of Disease, 11th ed., 2026, Ch. 12

Loeffler's endomyocarditis – eosinophils & mild myocardial necrosis



- Myocardial biopsy, H&E stained, with **eosinophilic infiltration** (arrows) of tissue (low power light microscopy left, high power right)

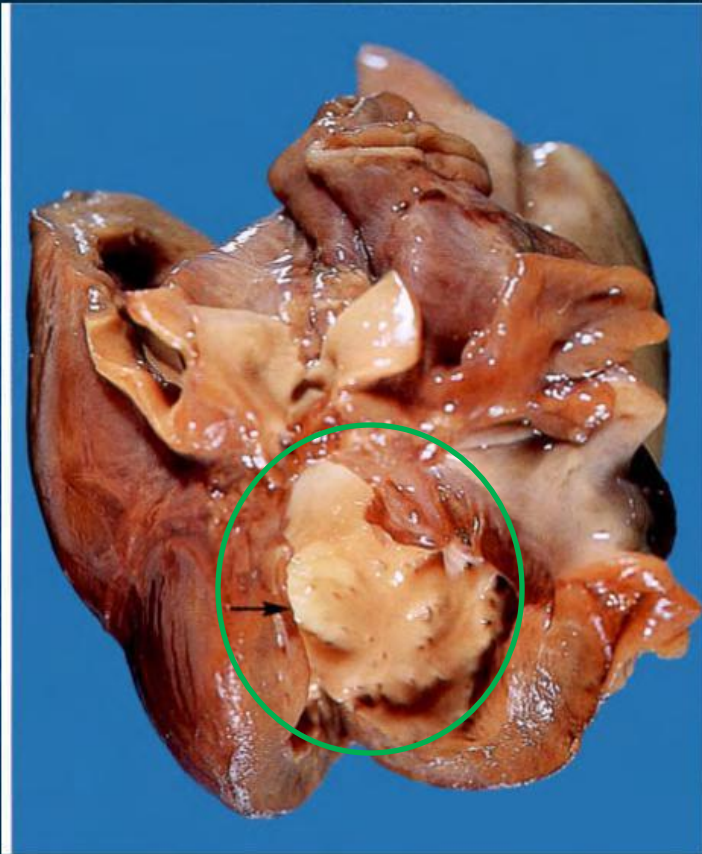
Reiko Ito, Takuma Sakon, Kiyoshi Yasui, Mari Iwamoto and Yukoh Aihara
Allergology International 2003; 52: 219-224

Endocardial fibroelastosis

- **Uncommon**, obscure etiology
- Focal or diffuse **fibroelastic thickening** usually of **mural LV endocardium**
- Usually in the **first 2 years of life**
- 1/3rd of cases, accompanied by **aortic valve obstruction** or a congenital cardiac anomaly
- Might be common morphologic **endpoint of several different insults?** including viral infections (e.g., intrauterine exposure to mumps) or mutations in the gene for **tafazzin**, which affects mitochondrial inner membrane integrity

Source: Robbins, Cotran & Kumar Pathologic Basis of Disease, 11th ed., 2026, Ch. 12

Left ventricle of an infant's heart with endocardial fibroelastosis who died of cardiac failure



Fibroelastosis of the left ventricle.

M/3 weeks. Death from cardiac failure. The endocardial surface of the left ventricle is lined by a thick layer of white tissue.

- The endocardium is thick and opaque due to proliferation of **collagen & elastic fibers**
- Etiology unknown

30

<http://zavantag.com/docs/2110/index-46183.html>

Cardiomyopathy compare & contrast

	Dilated	Hypertrophic	Restrictive
Ventricular morphology	Dilated LV with possibly some myocardial thickening	Marked hypertrophy often asymmetric	Fibrotic or infiltrated myocardium
Symptoms	Fatigue, weakness, dyspnea, orthopnea , PND	Dyspnea, angina , syncope	Dyspnea, fatigue
Physical Exam	Pulmonary rales, S3 ; if RV failure present, JVD, hepatomegaly, peripheral edema	S4 ; if outflow obstruction present, systolic murmur loudest at left sternal border, may be accompanied by mitral regurgitation	Signs of RV failure: JVD, hepatomegaly, peripheral edema
Pathophysiology	Impaired systolic contraction	Impaired diastolic relaxation; LV systolic function vigorous , often with dynamic obstruction	"Stiff" LV with impaired diastolic relaxation but normal systolic function

Thank you,
Lecture feedback form

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>